

CS208 Cloud Computing and Cloud Networking

Assignment 3: Create your own Cloud

(Due 11:59 pm, 14 Feb 2025)



Openstack

This tutorial will walk you through the process of creating a small cloud on CloudLab using OpenStack. Your copy of OpenStack will run on bare-metal machines that are dedicated for your use for the duration of your experiment. You will have complete administrative access to these machines, meaning that you have full ability to customize and/or configure your installation of OpenStack.

General Instructions:

- Follow the steps shown in the link [1] from “14.4 Building Your Own OpenStack Cloud” onwards on Cloudlab and document your entire progress with screenshots and relevant information. Your documentation shall be submitted to iLearn. Instead of
- instantiating **one virtual machine** as shown in the document [1], instantiate **three virtual machines** on the **same physical node**. **Note:** These virtual machines are
- created from Openstack dashboard not from Cloudlab dashboard.

[1] Cloudlab Openstack installation: <https://docs.cloudlab.us/openstack-tutorial.html>

[2] <http://docs.cloudlab.us/virtual-machines-advanced.html>

Answer the following questions

1. In the context of cloudlab, what are the differences between virtual and physical nodes?
2. How is destination network-address translation (DNAT) different from source network-address translation (SNAT)?
3. Furthermore, using screenshots depict your ability to ping a virtual machine from another virtual machine on the same node. Using the ping utility, estimate the round trip time between two virtual machines.

Grading

What is tested?

- You have to create a single Word or PDF document for answers and put your screenshots.

Grade Distribution

- Answers - 20 points
- Openstack Installation steps - 50 points
- Ping virtual machine from another virtual machine - 30 points